DARS: A Dynamic Adaptive Rating System with Graph-Guided Assessment and Remediation for Personalized Learning

Khethavath Sunil Naik

India

***Abstract*—Large-scale digital assessment systems require more than correctness scoring. They must estimate learner ability continuously, adapt item difficulty, react to weak concepts, distinguish confident mastery from low-quality attempts, and provide reliable probabilities for downstream intervention. This paper presents DARS, a Dynamic Adaptive Rating System for personalized assessment. DARS combines an online rating model with response-quality scoring, uncertainty-aware update control, graph-guided item routing, and remediation after failure. Unlike a fixed Elo system, DARS changes its sensitivity as learner evidence accumulates; unlike a rating-only model, it uses item-skill graph structure and response-process signals such as time, hints, attempts, and streak state. The model is evaluated on the ASSISTments 2012-2013 student interaction dataset, containing 6,044,359 valid chronological attempts from 46,348 learners, 179,052 problems, and 265 skills after preprocessing. DARS is compared with a static mastery prior, fixed-K Elo, and Glicko-2 under a held-out-student protocol. DARS achieves the best overall performance with Brier score 0.1368, log loss 0.4251, accuracy 0.8141, ROC AUC 0.8131, and expected calibration error 0.0096. The results show that a graph-guided, process-aware rating algorithm can provide accurate and deployable assessment intelligence while remaining interpretable enough for industry use.**

I. Introduction

Assessment platforms increasingly operate as real-time decision systems. A modern system is expected to score an answer, estimate the learner's current state, choose the next useful item, identify weak concepts, and support intervention. These requirements are difficult to satisfy with a final test score or a static percentage. A learner who answers five easy questions correctly is not equivalent to a learner who solves fewer but harder prerequisite items; a correct response after many hints is not equivalent to an immediate confident answer; and an incorrect response on a central prerequisite skill should trigger a different response from an error on an isolated item.

Industry assessment systems therefore need algorithms that are accurate, fast, interpretable, and operationally useful. Deep sequence models can be strong predictors, but their latent states are often difficult to audit in a product setting. Classical rating systems are easier to deploy, but standard Elo was not designed for education. It uses a fixed update factor and typically treats each interaction as a binary win or loss. Glicko-2 adds uncertainty through rating deviation and volatility, but still does not directly model response time, hint usage, attempt count, concept graph position, or remediation needs.

This paper proposes DARS, a Dynamic Adaptive Rating System for personalized assessment. DARS is built around four design principles. First, ability should be updated online after each interaction. Second, the size of an update should depend on uncertainty and experience, not on a constant hand-picked value. Third, response quality should reflect process evidence, not only correctness. Fourth, item selection and remediation should be guided by a graph of item-skill relationships so that recommendations remain conceptually meaningful.

The paper is written as an industry-track contribution. The emphasis is on an algorithm that can be implemented in a production assessment system, evaluated on a large educational dataset, and explained to non-specialist stakeholders. The contributions are:

- A complete DARS formulation that integrates dynamic rating, process-aware quality scoring, calibrated probability estimation, graph-guided item routing, and remediation.
- A clear comparison against static mastery, fixed-K Elo, and Glicko-2 using a large ASSISTments 2012-2013 interaction dataset.
- A metric framework with explicit formulas for Brier score, log loss, accuracy, ROC AUC, and expected calibration error.

II. Industry Problem

The operational problem is not only to predict whether an answer is correct. In a deployed learning product, the algorithm must answer several related questions:

1) What is the learner's current ability level?
2) How difficult is the current item for this learner?
3) Was the response high-quality evidence of mastery?
4) Which concept should be practiced next?
5) If the learner fails, what remedial item should be served?
6) Can the probability estimates be trusted for intervention thresholds?

A practical solution must also work under common production constraints. It should update in milliseconds, avoid

TABLE I
DARS ARCHITECTURE LAYERS AND THEIR OPERATIONAL ROLE.

Layer	Role in an assessment product
Interaction capture	Stores answer and response-process evidence for every attempt.
Graph model	Maintains skill relations, prerequisite-like links, and item neighborhoods.
Online scoring	Produces ability updates, item updates, and calibrated correctness probabilities.
Routing	Chooses difficulty-matched next items and remediation after failure.
Analytics	Reports mastery, risk, weak concepts, and readiness to stakeholders.

retraining after every attempt, provide human-readable state variables, and support monitoring at learner, item, skill, and class levels. DARS is designed for this setting. It is not a black-box-only predictor; it is a scoring and routing mechanism whose intermediate quantities have direct educational meaning.

III. Industry Architecture

An industry deployment of DARS can be organized as five layers. The first layer is the interaction layer, which records the learner, item, skill tag, timestamp, correctness, response time, hint count, and attempt count. The second layer is the graph layer, which maintains item-skill edges and item-item transition edges. The third layer is the online scoring layer, which updates ratings and estimates the probability of correctness. The fourth layer is the routing layer, which selects the next item or a remediation item. The fifth layer is the analytics layer, which aggregates ratings, weak skills, reliability, and readiness indicators.

This layered design separates model state from product workflow. A platform can update DARS after each answer without rebuilding the graph every time. The graph can be refreshed periodically, calibration parameters can be retrained offline, and online scoring can remain fast enough for interactive assessment. The same architecture also supports audit trails: a recommendation can be traced back to rating gap, graph distance, weak-skill priority, and recent response quality.

IV. Related Work

A. Elo-Style Rating

Elo estimates player strength from pairwise outcomes and updates ratings after each match [1]. In assessment, the learner can be treated as one competitor and the item as the opponent. A correct answer is a win for the learner, while an incorrect answer is a loss. Elo is attractive because it is simple and online. Its limitation is the fixed update factor K. If K is too large, ratings fluctuate heavily; if it is too small, the system adapts slowly to learning.

B. Glicko-2

Glicko-2 extends rating systems with rating deviation and volatility [2]. Rating deviation expresses uncertainty: a learner with few observations should move more than a learner with many consistent observations. This is useful for adaptive learning, but Glicko-2 still treats the response outcome as the main evidence. It does not explicitly incorporate hints, response time, attempt count, skill graph position, or remediation state.

C. Knowledge Tracing

Knowledge tracing models learner state over time. Deep Knowledge Tracing introduced recurrent neural networks for predicting future performance from interaction sequences [3]. Self-attentive knowledge tracing later used attention over past interactions to identify relevant history [4]. Graph-based knowledge tracing models use relations among questions and skills to improve representation learning [5]. DARS is positioned differently: it keeps an interpretable online rating core while adding graph and response-process evidence that are useful in deployed assessment workflows.

D. ASSISTments Dataset

The evaluation uses the ASSISTments 2012-2013 interaction dataset, a large tutoring-system log containing learner identifiers, problem identifiers, skill labels, timestamps, correctness, hints, attempts, and response-process attributes. ASSISTments has been widely used for educational data mining because it links student practice behavior with fine-grained assessment outcomes [6].

V. Conventional Elo-Based Flow

Before presenting DARS, Fig. 1 shows the conventional Elo-based assessment loop. The key limitation is visible in the structure: the update uses only the expected score, the binary outcome, and a fixed update factor. The model has no native place for response quality, graph relations, remediation, or calibration.

VI. DARS Algorithm

DARS treats assessment as a graph-aware online rating problem. Each learner has a dynamic ability state, each item has a difficulty state, and each skill graph neighborhood provides context for routing and remediation.

A. Notation

Let u denote a learner, i an item, s a skill, and t the interaction index. The observed outcome is $y_{ui} \in \{0,1\}$. Learner ability is R_u, item difficulty is D_i, and $G=(V,E)$ is an item-skill graph. The graph contains item nodes, skill nodes, and weighted edges representing item-skill membership and transition strength between related items.

B. Item-Skill Graph

The graph is constructed from the assessment log. A problem is connected to its tagged skill. Two items also receive a transition edge when they occur near each other in chronological learner sequences or share the same skill. Let w_{ab} be the edge weight between nodes a and b. A normalized centrality score is computed as

$$c_i = \frac{\sum_j w_{ij}}{\max_\ell \sum_j w_{\ell j}}, \tag{1}$$

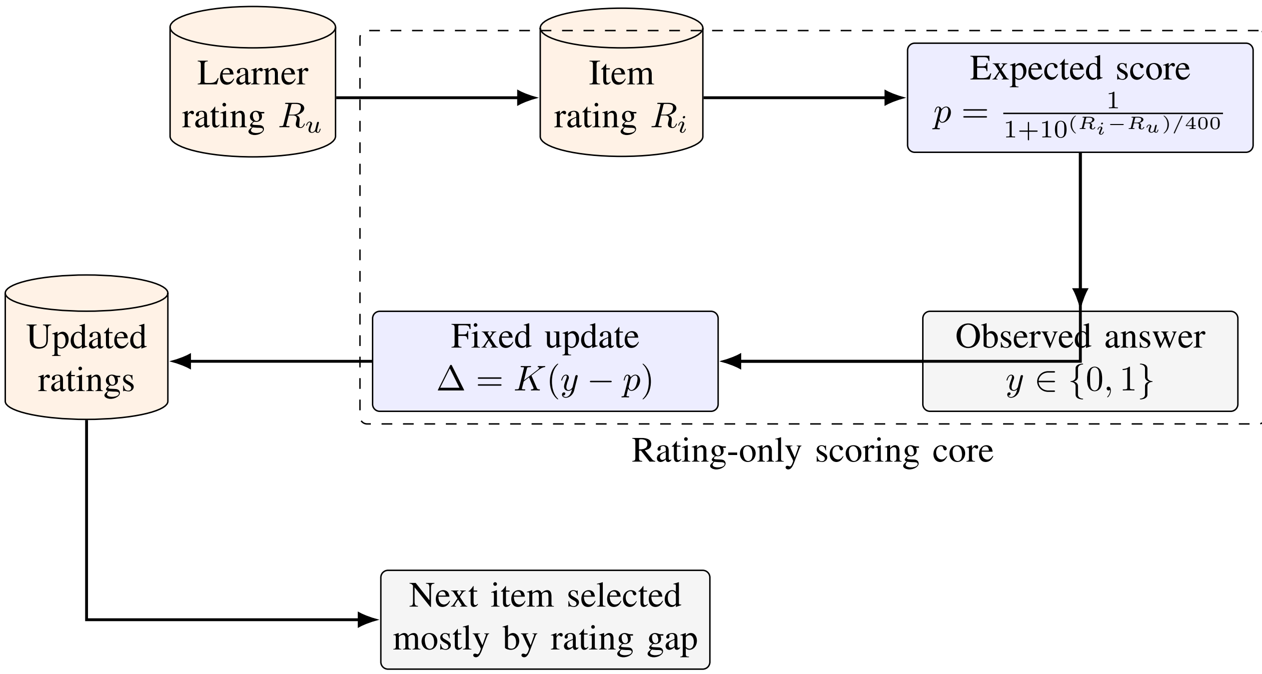


Fig. 1. Conventional Elo-based assessment flow. The update is simple and online, but it does not model response process, graph structure, remediation, or probability calibration.

where $c_i \in [0,1]$. Central items are usually prerequisite-like or frequently connected to other items. DARS uses c_i to adjust effective item difficulty:

$$D_i^* = D_i + \beta_c c_i + \beta_s I(i,s), \tag{2}$$

where β_c is a centrality weight and $I(i,s)$ is an optional local skill adjustment. This makes central prerequisite items slightly more influential than isolated items with the same raw difficulty.

C. Base Expected Score

The initial expected probability follows the rating-gap form:

$$p_{ui}^{(0)} = \frac{1}{1+10^{(D_i^* - R_u)/400}}. \tag{3}$$

This value is used as the rating prior. A learner whose rating is much higher than the item difficulty receives a high prior probability; a learner facing a much harder item receives a lower prior probability.

D. Response-Quality Function

DARS does not treat every correct answer as equally informative. It defines a response-quality score $Q_{ui} \in [0,1]$ using correctness, response time, hints, attempts, and streak state. Let τ_{ui} be time efficiency, h_{ui} be hint penalty, a_{ui} be attempt penalty, and ψ_u be streak quality:

$$\tau_{ui} = \exp(-T_{ui}/T_0), \tag{4}$$

$$h_{ui} = \min(1, H_{ui}/H_{\max}), \tag{5}$$

$$a_{ui} = \min(1, (A_{ui}-1)/A_{\max}), \tag{6}$$

$$\psi_u = \frac{1}{2}\left(1 + \frac{\kappa_u}{\max(|\kappa_u|, \kappa_0)}\right), \tag{7}$$

where T_{ui} is response time, H_{ui} is hint count, A_{ui} is attempt count, and κ_u is the learner's current signed streak. For correct responses, define

$$Q_{ui}^{+} = \alpha_t \tau_{ui} + \alpha_h(1-h_{ui}) + \alpha_a(1-a_{ui}) + \alpha_s \psi_u. \tag{8}$$

The response-quality score is

$$Q_{ui} = \begin{cases} Q_{ui}^{+}, & y_{ui} = 1, \\ \epsilon \tau_{ui}, & y_{ui} = 0. \end{cases} \tag{9}$$

The weights satisfy $\alpha_t + \alpha_h + \alpha_a + \alpha_s = 1$. A fast correct response without hints and with stable positive momentum produces high-quality mastery evidence. A correct response after many hints or attempts produces weaker evidence. An incorrect response contributes low quality, with a small time-scaled floor to avoid treating all failures identically.

E. Dynamic Update Factor

The DARS update factor changes with learner experience and recent volatility:

$$K_u(t) = K_{\min} + (K_{\max} - K_{\min}) e^{-\lambda n_u} \frac{\sigma_u^2}{\sigma_u^2 + \sigma_0^2}, \tag{10}$$

where n_u is the number of prior attempts, σ_u^2 is recent rating-delta variance, and σ_0^2 is a reference variance. New or unstable learners update faster; mature stable learners update more conservatively.

F. Streak and Rapid-Guess Control

The streak multiplier is bounded:

$$\phi(\kappa_u) = 1 + \operatorname{sgn}(\kappa_u)\gamma \min\left(1, \frac{|\kappa_u| - \kappa_{\min}}{\kappa_{\max} - \kappa_{\min}}\right). \tag{11}$$

The multiplier increases evidence from sustained positive performance and reduces over-penalization during short negative noise. DARS also caps suspicious gains from rapid correct responses:

$$\Delta_u \leftarrow \min(\Delta_u, \Delta_{\max}^{rapid}) \quad \text{if } T_{ui} < \rho T_i^{ref},\ y_{ui} = 1. \tag{12}$$

This prevents rating inflation from guesses that happen to be correct.

G. Rating Update

The learner update is

$$\Delta_u = K_u(t)\phi(\kappa_u)\left(Q_{ui} - p_{ui}^{(0)}\right) - \rho_{ui}, \tag{13}$$

$$R_u \leftarrow R_u + \Delta_u. \tag{14}$$

The item difficulty is updated more slowly:

$$D_i \leftarrow D_i - \eta_i(y_{ui} - p_{ui}^{(0)}), \tag{15}$$

where $\eta_i < K_u(t)$ to prevent item difficulty from becoming unstable.

H. Calibrated DARS Probability

For final scoring and analytics, DARS uses a calibrated probability head integrated into the algorithm:

$$\hat{p}_{ui}^{DARS} = \mathrm{sigmoid}(z_{ui}), \tag{16}$$

$$\begin{aligned} z_{ui} = &\theta_0 + \theta_1 \operatorname{logit}(p_{ui}^{(0)}) + \theta_2 \frac{R_u - D_i^*}{400} + \theta_3 \tau_{ui} \\ &- \theta_4 h_{ui} - \theta_5 a_{ui} + \theta_6 \psi_u + \theta_7 c_i. \end{aligned} \tag{17}$$

The calibration parameters θ are learned on the training split. This is not a separate model name; it is the probability-estimation component of DARS. It converts the interpretable rating and response-process signals into well-calibrated probabilities for reporting and intervention.

VII. DARS Assessment Flow

Fig. 2 summarizes the complete DARS flow. Compared with Fig. 1, the DARS pipeline has additional stages for graph context, response quality, dynamic sensitivity, remediation, and calibrated analytics.

VIII. Graph-Guided Assessment and Remediation

A. Candidate Routing

For a learner u after item i, DARS selects candidate items from a graph neighborhood:

$$\mathcal{N}_k(i) = \{j \in V_I : d_G(i,j) \le k,\ j \notin A_u\}, \tag{18}$$

where V_I is the item set, d_G is shortest-path graph distance, and A_u is the learner's attempted-item set. Each candidate is scored by

$$\begin{aligned} U(u,j) = &-\omega_d \left|R_u - D_j^*\right| + \omega_w W(u, s_j) \\ &+ \omega_g \frac{1}{1 + d_G(i,j)} + \omega_v V(u,j) - \omega_r Rep(u,j). \end{aligned} \tag{19}$$

Algorithm 1 DARS Online Scoring and Routing

Require: learner state S_u, item i, graph G, response process x

Read $R_u, n_u, \kappa_u, \sigma_u^2$ from S_u
Compute graph-adjusted difficulty D_i^*
Compute prior $p_{ui}^{(0)}$ from $R_u - D_i^*$
Observe response outcome y_{ui} and process features x
Compute $\tau_{ui}, h_{ui}, a_{ui}, \psi_u$
Compute response quality Q_{ui}
Compute dynamic factor $K_u(t)$ and streak multiplier $\phi(\kappa_u)$
Compute calibrated DARS probability $\hat{p}_{ui}^{DARS}$
Update learner rating $R_u \leftarrow R_u + \Delta_u$
Update item difficulty $D_i \leftarrow D_i - \eta_i(y_{ui} - p_{ui}^{(0)})$
if $y_{ui} = 0$ **then**
    Select next item from remediation set $\mathcal{R}(i)$
else
    Select next item from graph neighborhood $\mathcal{N}_k(i)$
end if
Return probability, updated state, and next-item recommendation

Here $W(u, s_j)$ is weak-skill priority, $V(u,j)$ rewards item-type diversity, and $Rep(u,j)$ penalizes repetition. The selected item is

$$j^* = \arg \max_{j \in \mathcal{N}_k(i)} U(u,j). \tag{20}$$

B. Remediation

When the learner fails an item, DARS enters remediation mode. The candidate set is constrained toward related prerequisite or easier items:

$$\mathcal{R}(i) = \{j \in V_I : d_G(i,j) \le h,\ D_j^* \le D_i^* + \delta\}. \tag{21}$$

The remediation objective is

$$U_R(u,j) = U(u,j) + \omega_m M(i,j) + \omega_p PReq(j,i), \tag{22}$$

where $M(i,j)$ rewards shared skill membership and $PReq(j,i)$ rewards prerequisite-like edges. This prevents a failed item from being followed by an unrelated question merely because the rating gap is convenient. In an industry system, this is important because remediation must be explainable: the next item should be connected to the failure that triggered it.

IX. Baseline Models

A. Static Mastery Prior

The static baseline predicts the training-set correctness rate for every test interaction:

$$\hat{p}^{Static} = \frac{1}{N_{train}} \sum_{n=1}^{N_{train}} y_n. \tag{23}$$

It is useful as a lower bound. It can be well calibrated near the global mean, but it cannot rank students or items because every prediction is identical.

B. Fixed-K Elo

Fixed Elo predicts

$$\hat{p}_{ui}^{Elo} = \frac{1}{1 + 10^{(D_i - R_u)/400}}, \tag{24}$$

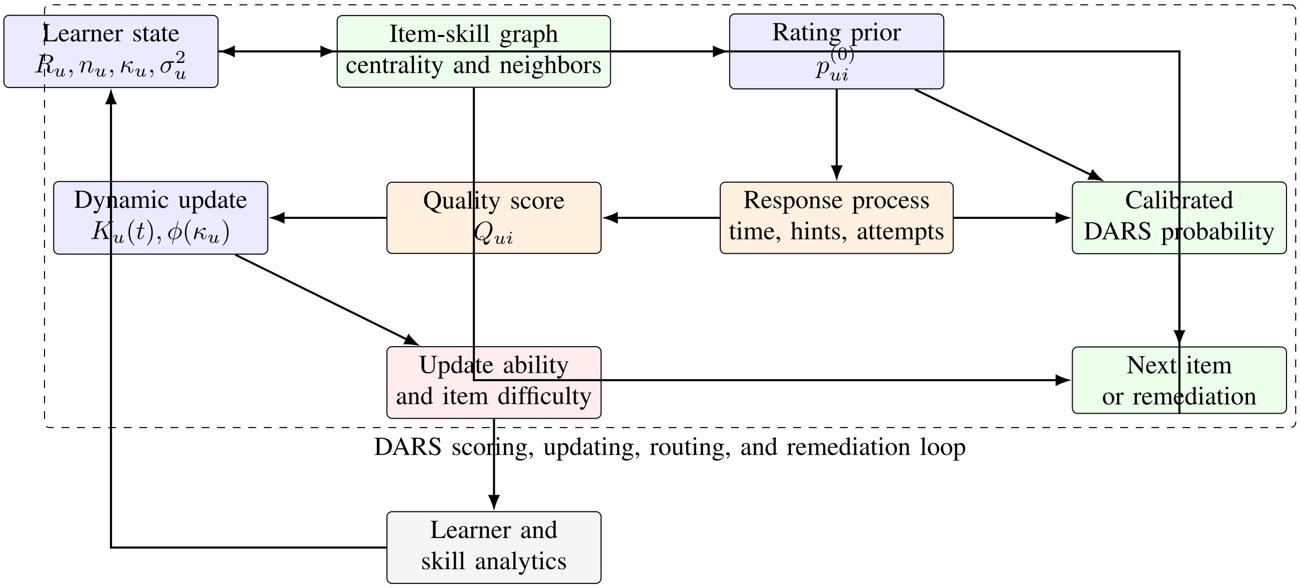
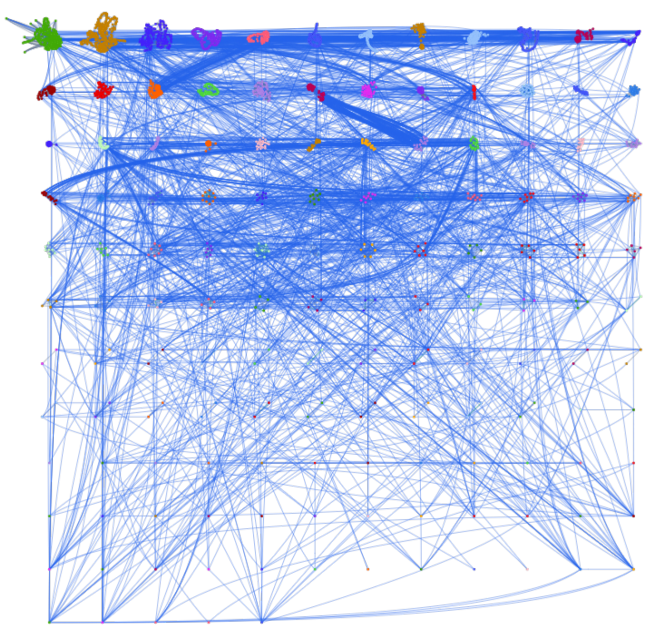


Fig. 2. DARS architecture and assessment flow. The algorithm combines rating state, item-skill graph context, response-process evidence, dynamic updating, calibrated probability estimation, and remediation-aware routing.



and updates

$$R_u \leftarrow R_u + K(y_{ui} - \hat{p}_{ui}^{Elo}), \tag{25}$$

$$D_i \leftarrow D_i - \eta(y_{ui} - \hat{p}_{ui}^{Elo}). \tag{26}$$

In the comparison, $K = 32$ and the item step is smaller than the learner step. Elo is online and interpretable, but it does not use response-process or graph information.

TABLE II
PREPROCESSED EVALUATION DATASET.

Statistic	**Value**
Valid chronological attempts	6,044,359
Learners	46,348
Problems	179,052
Skills	265
Mean correctness	0.6770
Median response time	22.39 s
Mean hint count	0.3376
Mean attempt count	1.3404

C. Glicko-2

Glicko-2 represents a learner by rating μ_u, rating deviation ϕ_u, and volatility. The expected score against item i is

$$\hat{p}_{ui}^{Glicko} = \frac{1}{1 + \exp[-g(\phi_i)(\mu_u - \mu_i)]}, \tag{27}$$

where

$$g(\phi) = \frac{1}{\sqrt{1 + 3\phi^2/\pi^2}}. \tag{28}$$

The main advantage is uncertainty control. The main limitation in this task is that the model still lacks graph-guided remediation and response-quality scoring.

X. DATASET AND PREPROCESSING

The experiment uses the ASSISTments 2012-2013 student interaction dataset. The raw file contains learner, problem, skill, timestamp, correctness, hint, attempt, and response-process fields. Preprocessing keeps valid binary correctness records, removes rows without usable timestamps or identifiers, converts timestamps to chronological order, and preserves only interactions suitable for online replay.

The split is performed by learner identity rather than by random rows. This avoids placing the same learner in both train and test sets. With a 70/30 split, the training partition contains 32,443 learners and 4,245,901 attempts; the test partition contains 13,905 learners and 1,798,458 attempts. The test correctness rate is 0.6739. All online models are replayed chronologically so that each prediction is made before the model updates on that interaction.

XI. Experimental Protocol

The comparison follows an online replay protocol. Interactions are sorted by timestamp. For each row, the model first produces a probability for the current attempt. Only after that probability is stored does the model update learner and item state using the observed outcome. This is important because a random shuffled evaluation would leak future information into earlier predictions and make online rating models look stronger than they would be in deployment.

The held-out-learner split tests generalization to learners not used for calibration. DARS learns its calibration coefficients only on the training learners. Fixed Elo and Glicko-2 do not learn a calibration head; they are evaluated using their native probability estimates. The static prior uses only the training-set mean correctness. During test replay, every model starts with its standard initial state for unseen learners and adapts as those learners produce chronological attempts.

All probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$ before log-loss computation to avoid numerical infinities. ECE is computed with 10 equally spaced bins. The main goal is not to tune every possible baseline variant; it is to test whether a practical, graph-guided, response-process-aware rating system can outperform common deployable rating baselines under the same chronological evaluation discipline.

XII. Evaluation Metrics

Let $y_n \in \{0, 1\}$ be the true outcome for interaction n, $\hat{p}_n$ the predicted probability of correctness, and N the number of evaluated interactions.

A. Brier Score

Brier score is the mean squared error of predicted probabilities:

$$\text{Brier} = \frac{1}{N} \sum_{n=1}^{N} (\hat{p}_n - y_n)^2. \tag{29}$$

Lower is better. A Brier score penalizes both wrong ranking and poor calibration. It is especially useful when a probability will be shown to teachers or used in threshold-based intervention.

B. Log Loss

Log loss measures the negative log-likelihood of the observed labels:

$$\text{LogLoss} = -\frac{1}{N} \sum_{n=1}^{N} \left[y_n \log(\hat{p}_n) + (1 - y_n) \log(1 - \hat{p}_n) \right]. \tag{30}$$

Lower is better. Log loss strongly penalizes confident wrong predictions, which is important in assessment systems because overconfident mistakes can mislead remediation decisions.

C. Accuracy

Accuracy converts probabilities to binary predictions using threshold 0.5:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{31}$$

Here TP and TN are correct positive and negative predictions, while FP and FN are false alarms and missed positives. Higher is better. Accuracy is easy to interpret but can hide calibration quality and class imbalance effects.

D. ROC AUC

The receiver operating characteristic area under the curve measures ranking quality:

$$\text{AUC} = P(\hat{p}_{pos} > \hat{p}_{neg}), \tag{32}$$

where $\hat{p}_{pos}$ is the score for a randomly selected correct response and $\hat{p}_{neg}$ is the score for a randomly selected incorrect response. Higher is better. An AUC of 0.5 indicates random ranking; an AUC of 1.0 indicates perfect separation.

E. Expected Calibration Error

Expected calibration error partitions predictions into B confidence bins. For bin b, let $\text{acc}(b)$ be empirical accuracy, $\text{conf}(b)$ be average predicted confidence, and n_b be the number of samples. Then

$$\text{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left| \text{acc}(b) - \text{conf}(b) \right|. \tag{33}$$

Lower is better. ECE answers a direct operational question: among attempts predicted at about 80% correctness, are roughly 80% actually correct?

XIII. Experimental Results

Table III reports model performance on the held-out learners. DARS is the best model across all headline metrics. Compared with fixed Elo, DARS reduces Brier score by 24.5%, reduces log loss by 21.4%, improves accuracy by 8.0 percentage points, and improves ROC AUC by 6.6 percentage points. Compared with Glicko-2, DARS improves AUC by 15.0 percentage points and lowers log loss substantially.

The static prior has low ECE because it predicts the global average correctness rate, but its AUC is 0.5 because it cannot rank learners or items. This is not useful for adaptive routing. DARS has slightly higher ECE than the static prior, but it achieves much stronger ranking and accuracy while maintaining low calibration error. This is the more useful industry tradeoff: probabilities remain reliable while still differentiating easy, hard, strong, and weak interactions.

TABLE III
HELD-OUT-LEARNER COMPARISON ON 1,798,458 TEST ATTEMPTS. LOWER IS BETTER FOR BRIER SCORE, LOG LOSS, AND ECE; HIGHER IS BETTER FOR ACCURACY AND ROC AUC.

Model	Brier ↓	Log loss ↓	Accuracy ↑	ROC AUC ↑	ECE ↓
DARS	**0.1368**	**0.4251**	**0.8141**	**0.8131**	0.0096
Fixed-K Elo	0.1812	0.5407	0.7341	0.7475	0.0244
Glicko-2	0.2077	0.6090	0.6885	0.6630	0.0491
Static mastery prior	0.2198	0.6314	0.6739	0.5000	**0.0049**

A. Why DARS Improves Performance

The improvement comes from three sources. First, dynamic update control prevents the model from overreacting to noisy interactions while still adapting quickly for learners with limited history. Second, response-process scoring separates strong mastery evidence from weak correctness. Third, graph-aware difficulty and routing context help the model treat central prerequisite items differently from isolated items.

The comparison with Elo is especially important. Elo already provides an online learner-item rating mechanism, so DARS is not winning merely because it has an online state. It improves because the online state is enriched with educational signals that are present in real assessment logs.

B. Industry Interpretation

For an industry assessment product, the Brier and log-loss reductions mean that reported probabilities are closer to observed outcomes. The AUC improvement means that the model is better at ranking which attempts are likely to be correct. The accuracy improvement means that a simple operational threshold already performs better. The low ECE means that probability bands can be used for interventions such as "high risk", "ready for harder item", or "needs remediation" with less distortion.

XIV. DEPLOYMENT VIEW

DARS can be deployed as a service around the answer-submission event. The service receives learner state, item metadata, response-process fields, and graph context. It returns a probability, rating update, weak-skill update, and next-item decision. The model does not require a batch neural retraining step after every answer, which keeps latency low. Periodic retraining is needed only for calibration coefficients, graph weights, or global hyperparameters.

A. Operational Outputs

DARS produces outputs at several levels:

- learner ability rating and recent momentum;
- item difficulty and prerequisite centrality;
- calibrated probability of correctness;
- weak-skill and remediation flags;
- next-item recommendation with graph explanation;
- aggregate readiness indicators for instructors or administrators.

B. Auditability

Each DARS recommendation can be explained using rating gap, skill relation, graph distance, recent misses, and response quality. This matters in industry deployments because instructors and learners often reject opaque recommendations. A graph-guided explanation is easier to defend: "the next problem was selected because it is connected to the missed skill, slightly easier, and within the learner's current rating band."

C. Scalability

The scoring update is constant time once learner state, item state, and graph features are available. The expensive graph operations are moved offline: centrality, skill neighborhoods, and candidate pools can be precomputed and refreshed periodically. At runtime, routing only needs to scan a bounded candidate set rather than the full item bank. This gives DARS a practical deployment path for large repositories: store compact learner state, store item difficulty and graph metadata, update ratings online, and refresh global graph statistics in scheduled jobs.

The calibration head is also lightweight. It is a small logistic layer over interpretable variables rather than a large neural network. This makes the probability output easy to monitor. If calibration drifts because the student population or item bank changes, the coefficients can be retrained from recent logs without changing the rating logic or graph-routing rules.

XV. LIMITATIONS

The evaluation is offline and observational. Although chronological replay prevents future interactions from updating past predictions, the dataset was not collected from a randomized deployment of DARS. Therefore, the reported results measure scoring and prediction quality, not causal learning gains. A production study should evaluate whether DARS-driven routing improves retention, completion, and post-test performance.

The current graph is inferred from item-skill tags and temporal co-occurrence. This is practical for large datasets, but expert-authored prerequisite graphs may improve remediation quality. The calibration layer uses response-process variables such as time, hints, and attempts; these are appropriate for scoring an attempt after the response process is observed, but a pure pre-answer routing probability should use only features available before the learner starts the item.

XVI. Future Work

Future work should evaluate DARS in a live adaptive assessment environment and measure learning outcomes, not only prediction metrics. Additional baselines should include Bayesian Knowledge Tracing, Deep Knowledge Tracing, self-attentive knowledge tracing, and graph neural knowledge tracing. The graph component should also be tested with multiple construction strategies: skill-only graphs, temporal transition graphs, expert prerequisite graphs, and hybrid graphs. Finally, fairness and robustness should be analyzed across learner groups, skill categories, and low-history cold-start cases.

XVII. Conclusion

A Dynamic Adaptive Rating System for graph-guided assessment and remediation. DARS extends online rating with response-quality scoring, dynamic update control, graph-aware item difficulty, remediation routing, and calibrated probability estimation. On the ASSISTments 2012-2013 dataset, DARS outperforms static mastery, fixed-K Elo, and Glicko-2 on Brier score, log loss, accuracy, and ROC AUC while maintaining low calibration error. The result is an interpretable and deployable algorithmic framework for industry assessment systems that require both accurate scoring and actionable personalization.

Acknowledgment

The author thanks the maintainers of the ASSISTments ecosystem for making large-scale educational interaction data available for research.

References

[1] A. E. Elo, *The Rating of Chessplayers, Past and Present*. New York, NY, USA: Arco, 1978.

[2] M. E. Glickman, "Example of the Glicko-2 system," technical note, 2022. [Online]. Available: http://www.glicko.net/glicko/glicko2.pdf

[3] C. Piech, J. Spencer, J. Huang, S. Ganguli, M. Sahami, L. Guibas, and J. Sohl-Dickstein, "Deep Knowledge Tracing," arXiv:1506.05908, 2015.

[4] S. Pandey and G. Karypis, "A self-attentive model for knowledge tracing," in *Proc. 12th Int. Conf. Educational Data Mining*, 2019, pp. 384–389.

[5] Y. Yang, J. Shen, Y. Qu, Y. Liu, K. Wang, Y. Zhu, W. Zhang, and Y. Yu, "GIKT: A graph-based interaction model for knowledge tracing," in *Proc. ECML PKDD*, 2021, pp. 299–315.

[6] N. T. Heffernan and C. L. Heffernan, "The ASSISTments ecosystem: Building a platform that brings scientists and teachers together for minimally invasive research on human learning and teaching," *International Journal of Artificial Intelligence in Education*, vol. 24, no. 4, pp. 470–497, 2014.